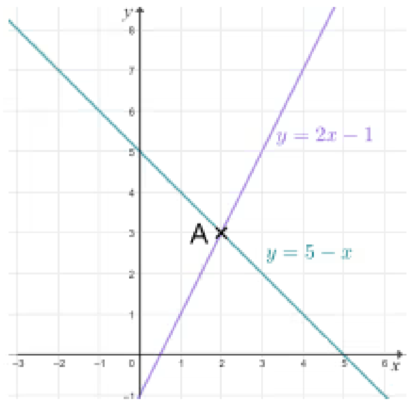
Name: ____________________________ Starter quiz

Problem solving with graphical representations

1 The diagram shows the lines $y = 2x - 1$ and $y = 5 - x$. The point A will help you find the solution to the ___________ equations $y = 2x - 1$ and $y = 5 - x$. Tick **1** correct answer



- [ ] intersecting
- [ ] non-linear
- [ ] quadratic
- [ ] simultanous

2 Sam plots the graphs of $y = 3x + 1$ and $y = 3x - 4$ on the same coordinate grid. How many points will lie on both lines? Fill in the blank

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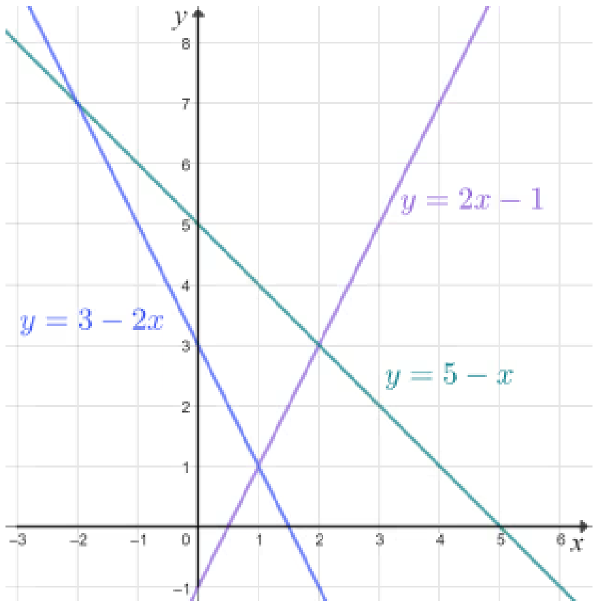
3 The linear equations $y = 2x - 9$ and $y = 3 - x$ intersect at the point with coordinates (4, -1). What is the solution to the equation $2x - 9 = 3 - x$? Tick **0** correct answers

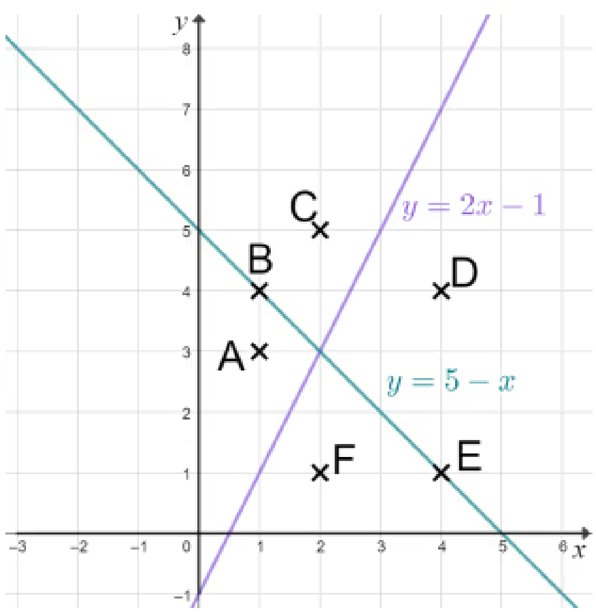
- [ ] $x = -1$
- [ ] $x = 4$
- [ ] $y = -1$
- [ ] $y = 4$

4 Use the graph to solve the simultaneous equations $y = 2x - 1$ and $y = 5 - x$. Tick **1** correct answer



- ☐ $x = 3, y = 2$
- ☐ $x = 7, y = -2$
- ☐ $x = 1, y = 1$
- ☐ $x = -2, y = 7$
- ☐ $x = 2, y = 3$

5 The diagram shows 6 points and the the lines $y = 2x - 1$ and $y = 5 - x$. Which points satisfy the inequality $y < 5 - x$? Tick **2** correct answers



- ☐ A
- ☐ C
- ☐ D
- ☐ F
- ☐ B and E

6 The diagram shows 6 points and the the lines $y = 2x - 1$ and $y = 5 - x$. The point that is the region $y > 5 - x$ and $y > 2x - 1$ is ____________________ . Fill in the blank